Wentian Zhu *Available May – August 2027* Email : Wentian.Zhu@uga.edu
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Education

- **University of Georgia** Athens, GA
  Ph.D. in Computer Science; Advisor: Prof. Le Guan *August 2024 – Present*
- **University of Science and Technology of China** Anhui, China
  M.S. in Cyberspace Science and Engineering; GPA: 4.0/4.3 *September 2022 – June 2024*
- **Huazhong University of Science and Technology** Hubei, China
  B.E. in Information Security; GPA: 3.7/4.0 *Sep 2018 – June 2022*

Experience

- **Samsung Research America** Mountain View, CA
  Research Engineer Intern *May 2026 - Aug 2026*
  - **AI Red-teaming**: Conducted Red-teaming for Agentic Browsers by designing attack methodologies targeting interactions between LLM Agents and browsers, generating proofs of concept for identified vulnerabilities, and proposing mitigation patches to improve system robustness and security.
  - **Agent Harness**: Designed and developed a suite of components for Agentic Browsers to improve task performance and execution efficiency, including a graph-based memory system with conflict resolution and incremental update capabilities, as well as on-device models that distill prerequisite information from agent requests to reduce context overhead. Contributed to the development of Monitor API by constructing golden datasets for evaluation and applying prompt engineering to improve performance.
  - **Agent Security Benchmarking**: Proposed a new scalable and extensible benchmark for evaluating the security of LLM agents, with fine-grained stepwise annotations and trajectory-level evaluation that localize security failures throughout agent execution. Leveraged the same structured evaluation design to train a detector-based defense that produces multidimensional judgments over agent behaviors, supporting more precise detection and attribution of security violations.
- **University of Georgia** Athens, GA
  Graduate Research Assistant *Aug 2024 - Present*
  - **Security Analysis of Agentic Browsers**: Conducted the first systematic security analysis of Agentic Browsers by evaluating indirect, implicit, and direct prompt-injection attacks across four commercial and open-source systems. Identified browser-specific attacks that exploit hidden content, malicious iframes, and hidden forms to manipulate agent actions, forge clicks, and leak private browser context; proposed visibility-aware, provenance-aware, and tool-layer defenses.
  - **Memory Security of LLM Agents**: Identified long-term memory as a new attack surface for the access control of LLM Agents and proposed FragFuse, a black-box attack that distributes prohibited intent across interactions and reconstructs it through memory retrieval. Evaluated the attack across four agent domains and multiple state-of-the-art guardrails, demonstrating substantially higher access-control bypass effectiveness than existing baselines while preserving the agents' ability to complete the original target tasks.
  - **Jailbreak Attacks**: Proposed MetaBreak, the first systematic jailbreak method against online LLM services, by exploiting LLM-inherent special tokens to manipulate chat templates and simultaneously bypass internal safety alignment, platform-added conversation wrappers, and external content moderation. Designed four attack primitives for response injection, turn masking, input segmentation, and semantic mimicry, and systematically evaluated them across four open-source LLMs and seven real-world chatbot and API services.
- **University of Science and Technology of China** Anhui, China
  Graduate Student *Sep 2022 - Jun 2024*
  - **Machine Learning Security**: Investigated model-leakage risks in mobile-deployed machine learning applications by reverse engineering native binaries to identify model-loading and decryption logic. Developed an automated leakage-detection framework that combines symbolic execution and data-flow analysis with cryptographic reasoning to trace sensitive model artifacts, recover protection workflows, and detect exploitable leakage.
  - **Privacy-Preserving Single Sign-On (SSO)**: Contributed to the design of a privacy-preserving SSO protocol. Formalized the protocol and verified its authentication and privacy properties under the Dolev–Yao model, complemented by game-based cryptographic security proofs and reduction arguments. Adapted the protocol to a browser-only deployment setting by integrating the Proof Key for Code Exchange (PKCE) flow.

- **ByteDance** Beijing, China
  Software Engineer Intern *May 2022 – Jul 2022*
  - **Backend Development**: Designed authentication logic for the Lark Task and added rich-text support. Developed Open Platform APIs and designed permission controls.
- **NetEase** Beijing, China
  Software Engineer Intern *Jul 2021 – Aug 2021*
  - **Plugin Development**: Performed reverse engineering on an instant-message application and developed plugins.

Publications

- **MetaBreak: Jailbreaking Online LLM Services via Special Token Manipulation**
  Wentian Zhu*, Zhen Xiang, Wei Niu, Le Guan
  Published in the 47th IEEE Symposium on Security and Privacy (IEEE S&P '26)
- **FragFuse: Bypassing Access Control of Large Language Model Agents via Memory-Based Query Fragmentation and Fusion**
  Wentian Zhu*, Zixin Rao*, Chan Lu, et al.
  To appear in the 35th USENIX Security Symposium (USENIX '26)
- **Whispers of Wealth: Red-Teaming Google's Agent Payments Protocol via Prompt Injection**
  Tanusree Debi*, **Wentian Zhu**, Pranjol Sen Gupta
  To appear in the International Conference on Intelligent Multimedia, Networking, and Security (IMNS '26)
- **When Browsers Think: A Systematic Security Check on Emerging Agentic Browsers**
  Wentian Zhu*, Dharma Tejaswini Janga, Haoyu He, et al.
  Under Review
- **Automatic Method to Analyze Deep Learning Model Leakage on Mobile Systems**
  Wentian Zhu*, Jingqiang Lin
  Published in Computer Application Research of China, No. 8, 2025
- **UPPRESSO: Untraceable and Unlinkable Privacy-PREserving Single Sign-On Services**
  Chengqian Guo*, **Wentian Zhu**, Jingqiang Lin, et al.
  Under Review

Skills

- **Programming**: Python, C/C++, Go, Java, TypeScript
- **Web Development**: Flask, Node.js, REST APIs
- **LLM Stack**: PyTorch, Transformers; vLLM, LiteLLM; Smolagents, LangChain, AutoGen
- **Program Analysis**: Reverse Engineering, Taint analysis, Data flow analysis